

CSCI2750 Data Mining & Information Processing

Lecture 7: The Support Vector Machine

Wai-Yiu Keung

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Department of Computer Science and Engineering
The Chinese University of Hong Kong

Linear Classification

- recall that linear classification is to find a function

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x})$$

where $\mathbf{w} \in \mathbb{R}^n$ is the model parameter

- the problem can be cast as finding $\mathbf{w}$ such that

$$\mathbf{w}^\top \mathbf{x}^{(i)} \geq 0 \text{ if } y^{(i)} = 1, \quad \mathbf{w}^\top \mathbf{x}^{(i)} < 0 \text{ if } y^{(i)} = -1,$$

for $i = 1, \dots, m$; this can be written as

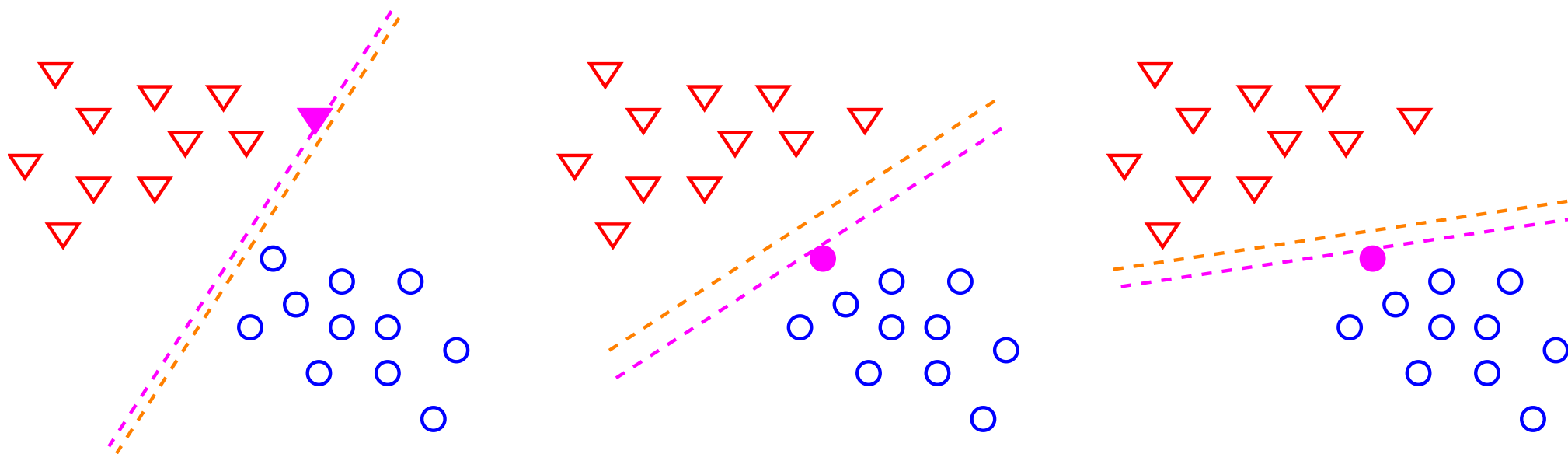
$$y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} > 0, \quad i = 1, \dots, m.$$

- in terms of optimization language, this is a feasibility problem:

$$\begin{array}{ll} \min_{\mathbf{w} \in \mathbb{R}^n} & 0 \\ \text{s.t.} & y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} > 0, \quad i = 1, \dots, m. \end{array}$$

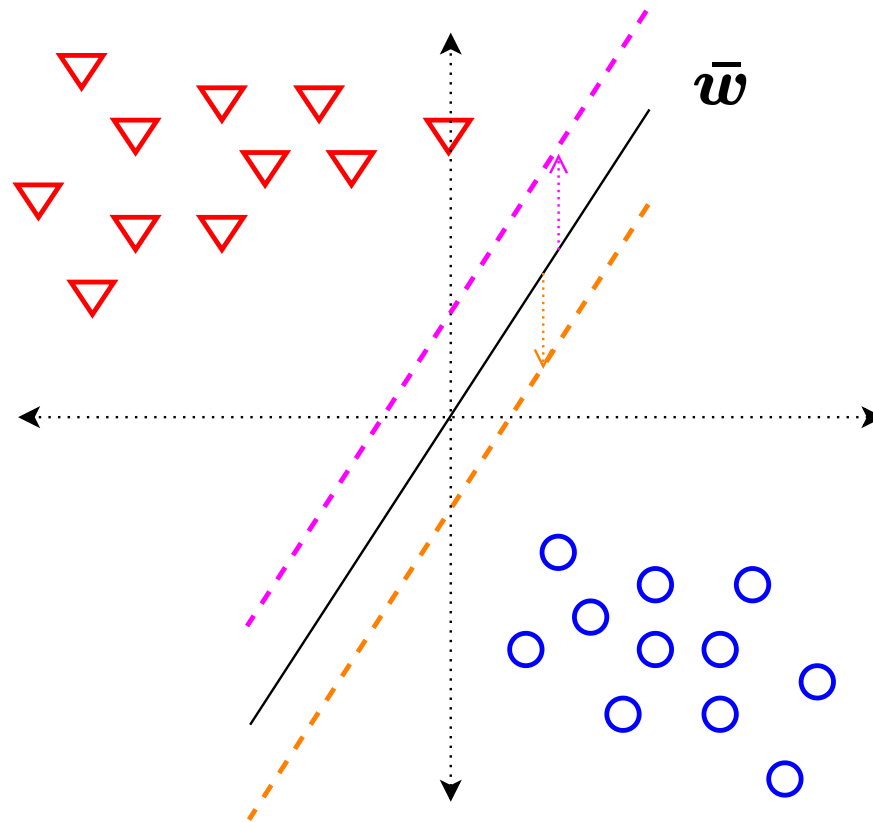
in which minimizing 0 means we have no objective but to satisfy the constraints, and we've managed to solve it by the perceptron algo.

Recall: Margin



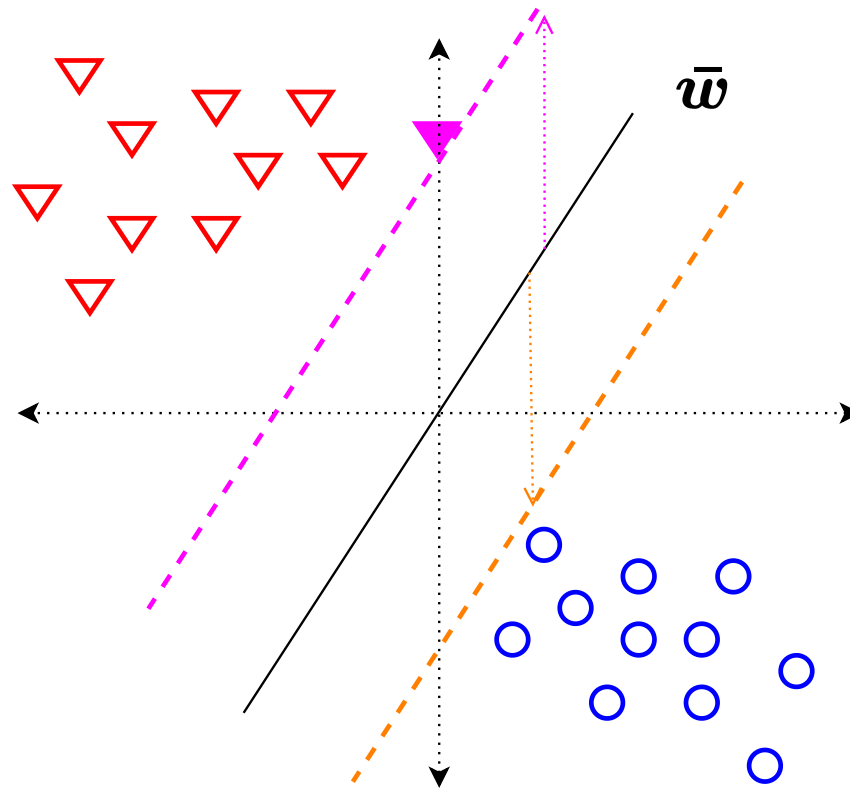
- recall that the margin is the **distance** between the **separation line** and the **nearest x**
- the physical meaning of margin γ is the width between the **orange** and **pink** dashed lines in the above figures
- This lecture: assume data are i) separable and ii) zero bias, we seek to **maximize the margin!**

Margin Maximization



- for any separation plane $\bar{w}$ that does the job, i.e., $y^{(i)} \bar{w}^\top \mathbf{x}^{(i)} > 0$ for all i
- consider two copies of it, one goes upward, another goes downward
- the copies keep going up or down until it hits one data point

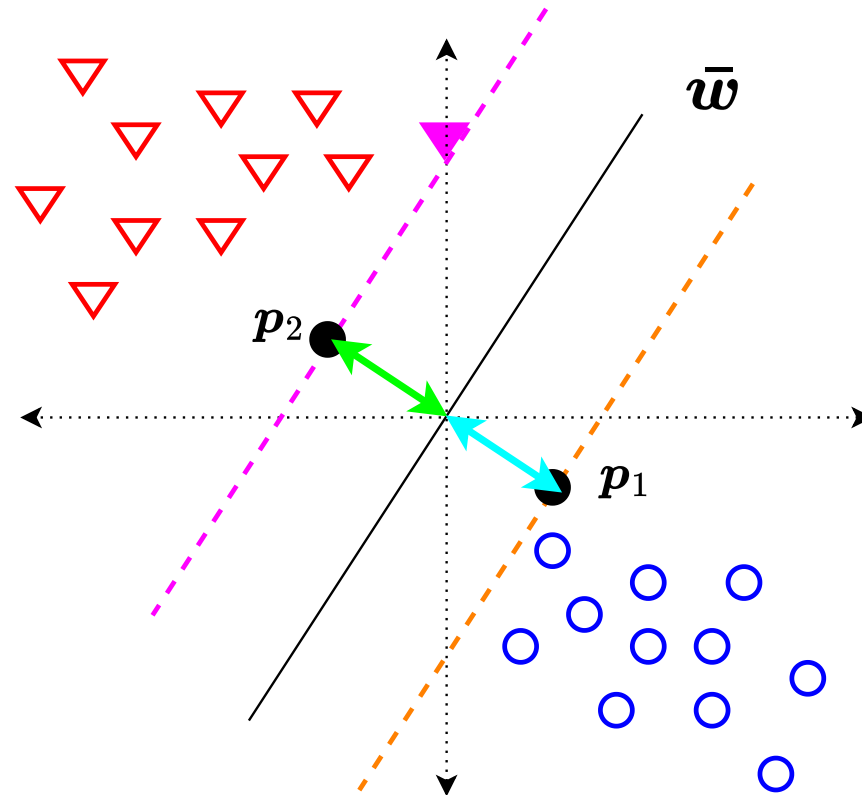
Margin Maximization



- if a point is hit, we see: the planes are $\bar{w}^\top x = b$, and $\bar{w}^\top x = -b$, for a $b > 0$
- note that the following must hold: $y^{(i)} \bar{w}^\top x^{(i)} \geq b$
- we can define $w = \bar{w}/b$ and obtain the inequality

$$y^{(i)} w^\top x^{(i)} \geq 1, \quad i = 1, 2, \dots, m.$$

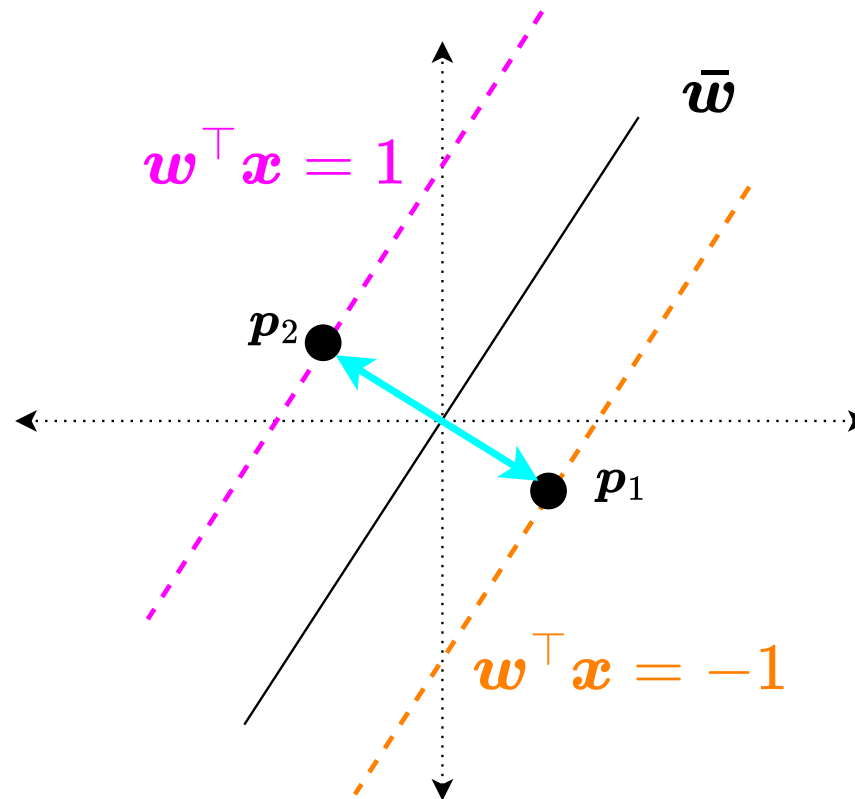
Margin Maximization



- for a given $\bar{w}$, the **margin** $\gamma(\bar{w})$ is exactly one-half of the dist. between the two planes
- let p_1 lies on the plane $\mathbf{w}^\top \mathbf{x} = -1$ and p_2 lies on the plane $\mathbf{w}^\top \mathbf{x} = 1$, then

$$\mathbf{w}^\top (\mathbf{p}_2 - \mathbf{p}_1) = 2$$

Margin Maximization



- since w and $(p_2 - p_1)$ are orthogonal to each other, we have

$$\|w^\top (p_2 - p_1)\|_2 = \|p_2 - p_1\|_2 \cdot \|w\|_2;$$

using the equation on last slide: $\|p_2 - p_1\|_2 = 2/\|w\|_2$.

- hence, the margin is $\gamma(\bar{w}) = \frac{\|p_2 - p_1\|_2}{2} = \frac{1}{\|w\|_2} \propto \frac{1}{\|w\|_2^2}$

Problem Formulation

- our target is to choose a separation plane $\bar{\mathbf{w}}$ such that the margin is maximized
- this can be formulated as

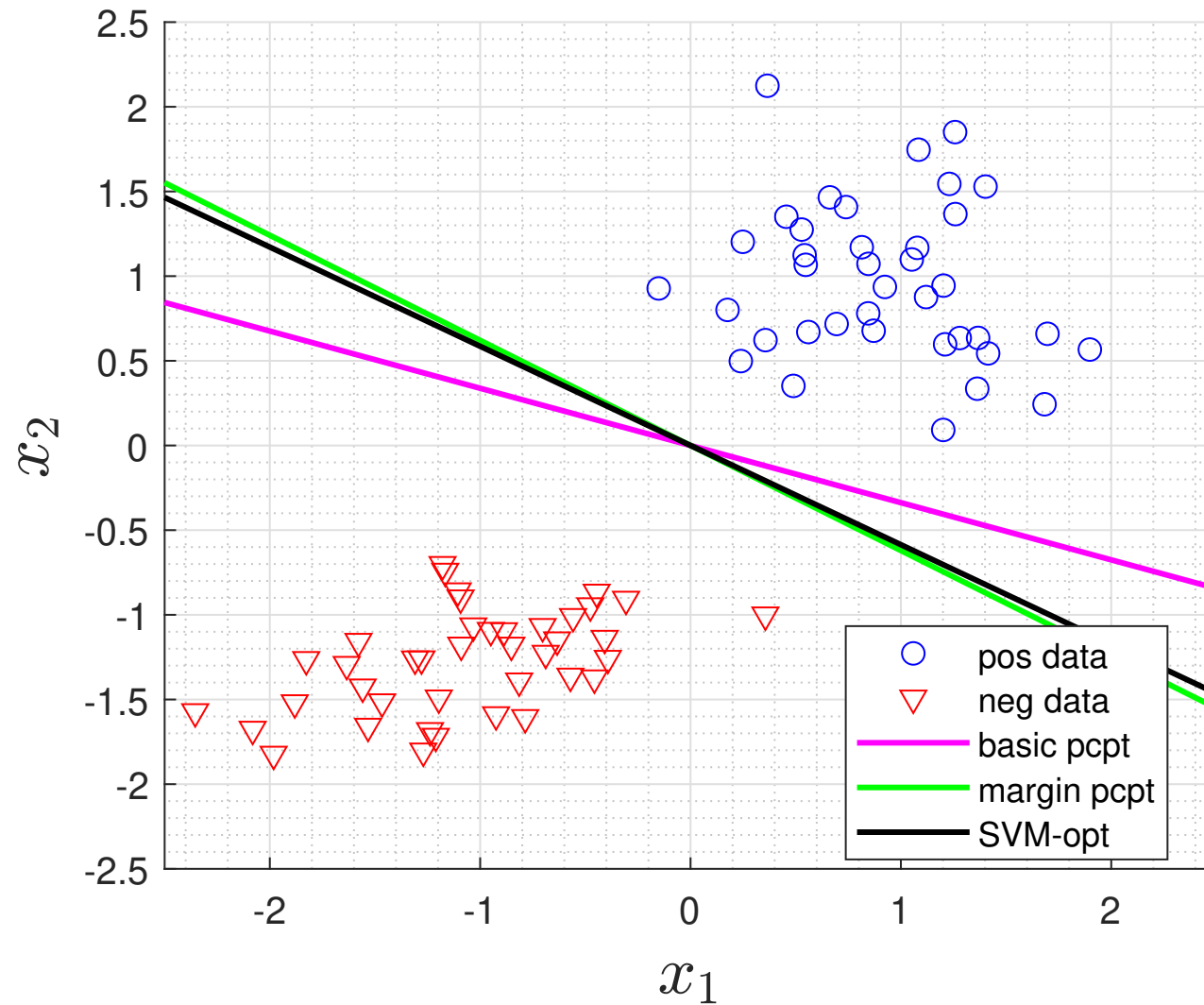
$$\begin{aligned} \max_{\bar{\mathbf{w}} \in \mathbb{R}^n, b > 0} \quad & \gamma(\bar{\mathbf{w}}) \\ \text{s.t.} \quad & y^{(i)} \cdot \bar{\mathbf{w}}^\top \mathbf{x}^{(i)} \geq b, \quad i = 1, \dots, m \end{aligned}$$

which can be simplified as

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^n} \quad & \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} \geq 1, \quad i = 1, \dots, m. \end{aligned}$$

- this is a [quadratic programme](#), which is convex and solvable by CVX

Example: Comparison of SVM, Perceptron, and Margin-Pcpt



Support Vector...?

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^n} \quad & \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} \geq 1, \quad i = 1, \dots, m. \end{aligned}$$

- for an optimal $\mathbf{w}_{\text{opt}}$, there exists at least one point that satisfies $y^{(i)} \mathbf{w}_{\text{opt}}^\top \mathbf{x}^{(i)} = 1$
- Proof: Suppose $\mathbf{w}$ is an optimal. If $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^m$ all has $y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} > 1$, then

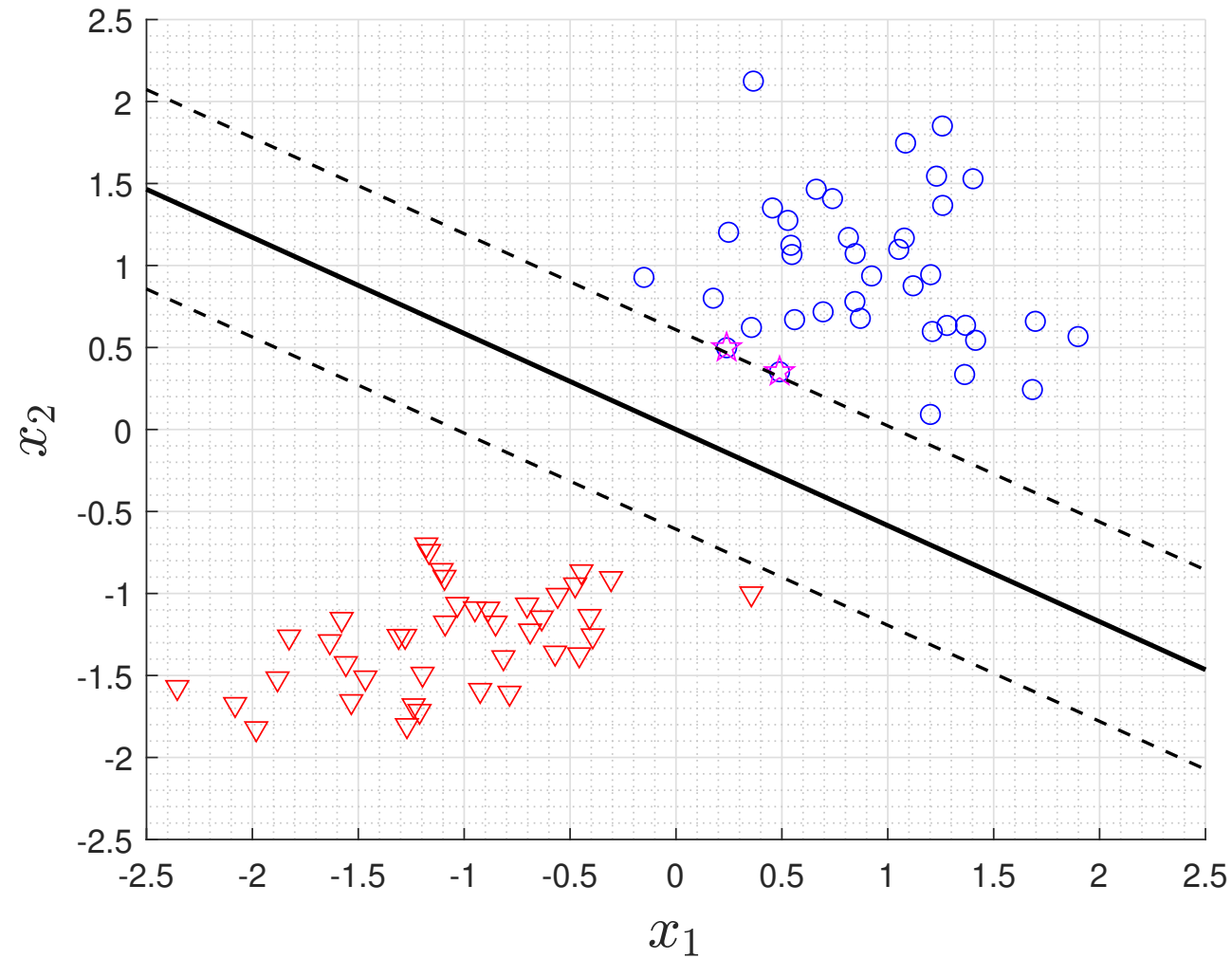
$$\mathbf{w}' = \mathbf{w}/C$$

where $C = \max_{i=1, \dots, m} \{|\mathbf{w}^\top \mathbf{x}^{(i)}|\}$ is also a feasible solution, but has smaller objective value.

This means $\mathbf{w}'$ is optimal instead of $\mathbf{w}$, implying a contradiction.

- the data points that satisfies $y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} = 1$ are referred to as the **support vectors**; they are what defines the **maximum margin of SVM**

Example: The Supporting Vectors



An Alternative Viewpoint: Robust Linear Classification

- consider the original linear separation problem

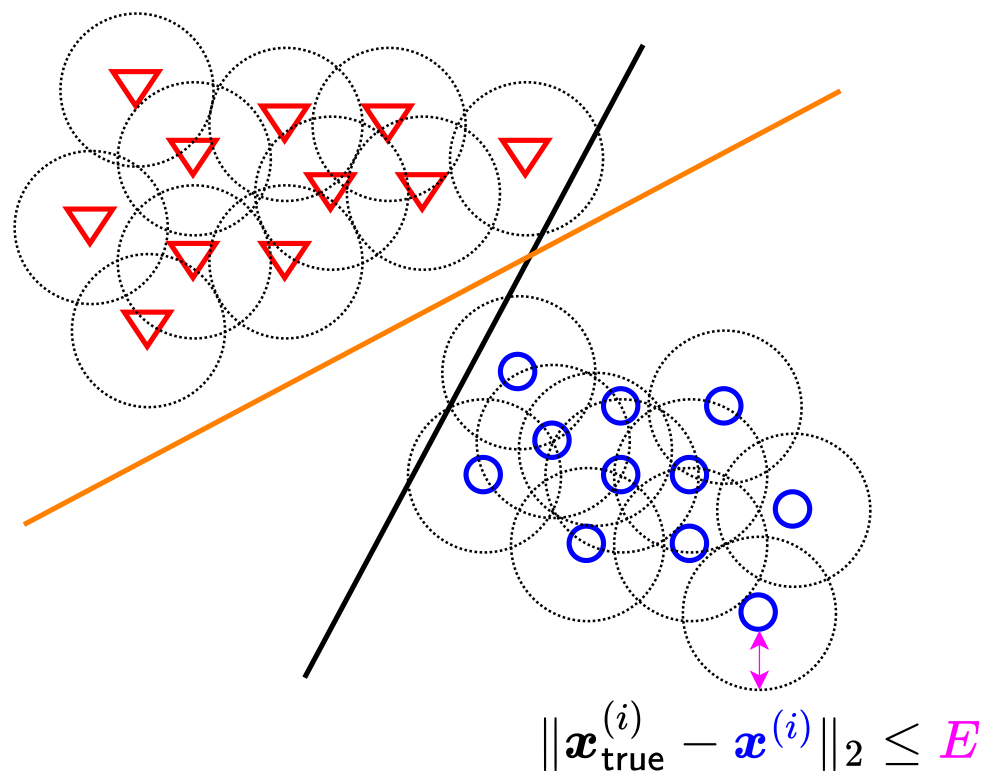
$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^n} \quad & 0 \\ \text{s.t.} \quad & y^{(i)} \cdot \mathbf{w}^\top \mathbf{x}^{(i)} > 0, \quad i = 1, \dots, m. \end{aligned}$$

- suppose that there are some inherent uncertainties in the data points $\{\mathbf{x}^{(i)}\}_{i=1}^m$, e.g., due to some data reception noise, or some modelling error
- uncertainty model:** we adopt the spherical uncertainty model:

$$\mathbf{x}_{\text{true}}^{(i)} = \underbrace{\mathbf{x}^{(i)}}_{\text{data}} + \underbrace{\mathbf{e}^{(i)}}_{\text{error}}, \quad \|\mathbf{e}^{(i)}\|_2 \leq E$$

in which $\mathbf{x}^{(i)}$ is the data point, $\mathbf{e}^{(i)}$ corresponds to an uncertainty vector, and $E \geq 0$ is the uncertainty level

An Alternative Viewpoint: Robust Linear Classification



- we wanted to maximize the uncertainty level E and separate the data
- this is formulated as

$$\begin{aligned} & \max_{\mathbf{w} \in \mathbb{R}^n, E \geq 0} && E \\ & \text{s.t.} && y^{(i)} \cdot \mathbf{w}^\top (\mathbf{x}^{(i)} + \mathbf{e}^{(i)}) \geq 0, \quad \text{for all } \|\mathbf{e}^{(i)}\|_2 \leq E, \\ & && i = 1, \dots, m. \end{aligned}$$

An Alternative Viewpoint: Robust Linear Classification

- denote $g_i(\mathbf{w}) = y^{(i)} \cdot \mathbf{w}^\top (\mathbf{x}^{(i)} + \mathbf{e}^{(i)})$, the problem is

$$\begin{aligned} \max_{\mathbf{w} \in \mathbb{R}^n, E \geq 0} \quad & E \\ \text{s.t.} \quad & g_i(\mathbf{w}) \geq 0, \quad \text{for all } \|\mathbf{e}^{(i)}\|_2 \leq E, \quad i = 1, \dots, m. \end{aligned}$$

- the function g can be written as

$$g_i(\mathbf{w}) = y^{(i)} \cdot \mathbf{w}^\top (\mathbf{x}^{(i)} + \mathbf{e}^{(i)}) = y^{(i)} \cdot [\mathbf{w}^\top \mathbf{x}^{(i)} + \mathbf{w}^\top \mathbf{e}^{(i)}]$$

and the lower bound of $g_i(\mathbf{w})$ among all $\|\mathbf{e}^{(i)}\| \leq E$ is

$$g_i(\mathbf{w}) \leq y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} - \|\mathbf{w}\|_2 E \iff y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} \geq \|\mathbf{w}\|_2 E$$

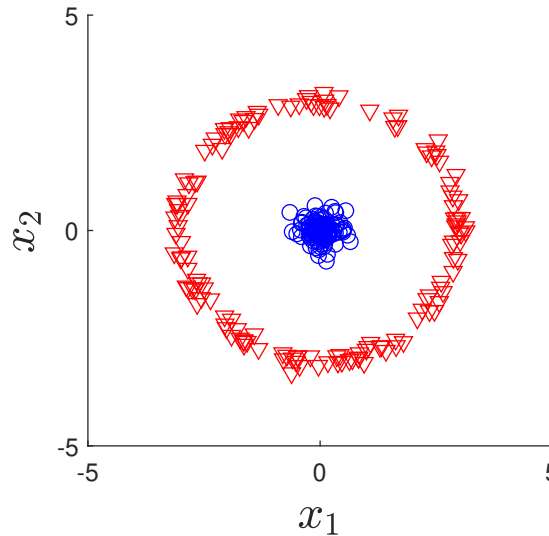
- w.o.l.g. we can assume $\|\mathbf{w}\|_2 = 1/E \implies E = 1/\|\mathbf{w}\|_2$ because $\mathbf{w}$ with any scaling won't hurt its optimality; hence we are able to reformulate

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^n} \quad & \|\mathbf{w}\|_2 \\ \text{s.t.} \quad & y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} \geq 1, \quad i = 1, \dots, m. \end{aligned}$$

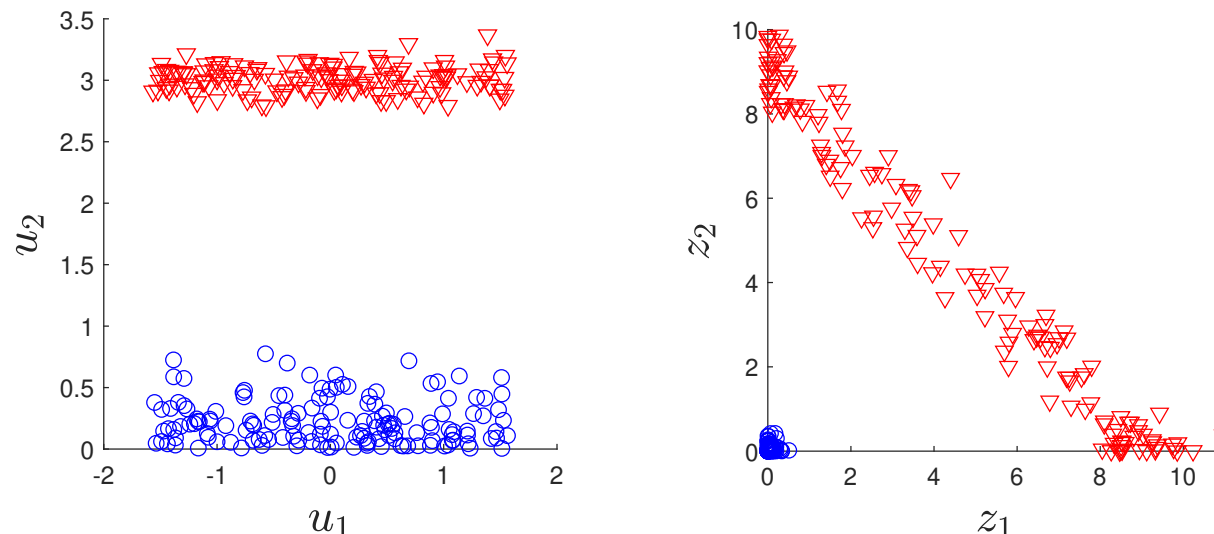
which is the the same quadratic programme we discussed above

Non-Linear SVM: Motivation

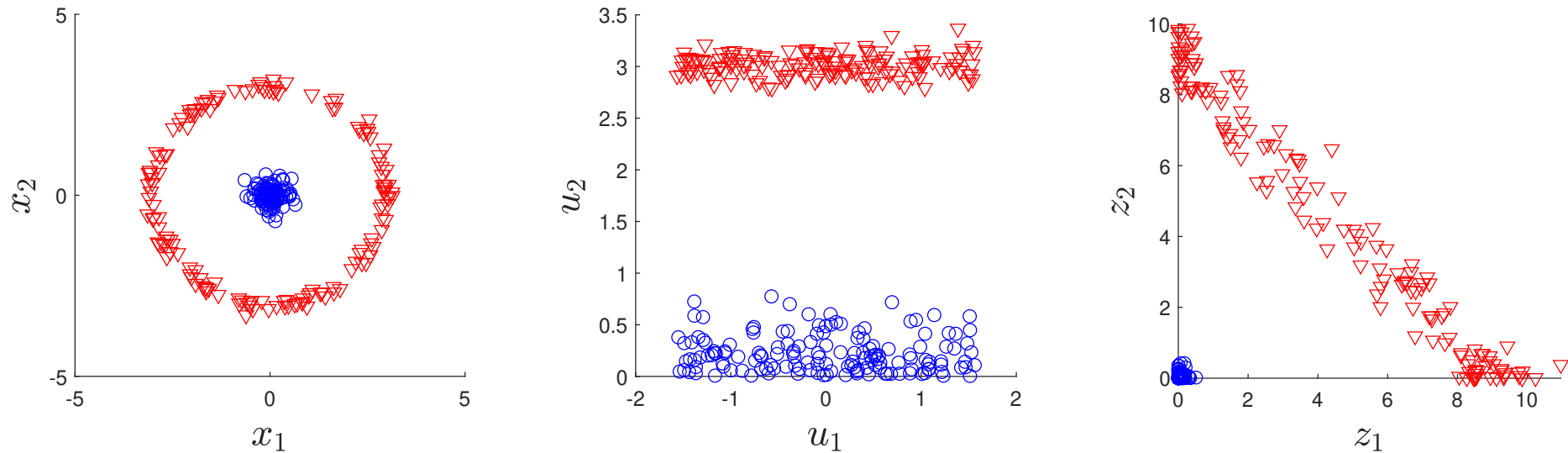
- so far we have been steering our discussions on the linearly separable case; but it is possible that the data is non-linearly separable, e.g.,



- we want to transform the data features s.t. it is linearly separable, e.g.,



Non-Linear SVM: Feature Mapping



- the nonlinear feature mappings we used are:

$$\phi_1(\mathbf{x}) = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} \tan^{-1}\left(\frac{x_2}{x_1}\right) \\ \sqrt{x_1^2 + x_2^2} \end{bmatrix} \quad \text{and} \quad \phi_2(\mathbf{x}) = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}$$

As seen from the above examples, the basis (u_1, u_2) and (z_1, z_2) are separable

- the same trick existed in polynomial regression wherein we apply linear regression on the mapping $\phi(x) = (x^0, x^1, x^2, \dots, x^{P-1})$, so that the model reads

$$h(x) = \mathbf{w}^\top \phi(x) = \sum_{p=0}^{P-1} w_p x^p$$

Non-Linear SVM: The Kernel

- given a feature mapping $\phi : \mathbb{R}^n \mapsto \mathbb{R}^\ell$, the non-linear SVM problem:

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^\ell} \quad & \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y^{(i)} \mathbf{w}^\top \phi^{(i)} \geq 1, \quad i = 1, \dots, m, \end{aligned}$$

where $\phi^{(i)} = \phi(\mathbf{x}^{(i)})$, can be readily solved; but in practice the dimension $\ell \gg n$ is large, and to calculate $\phi^{(i)}$ from $\mathbf{x}^{(i)}$ first and then solve SVM is inefficient

- kernel function:** given a nonlinear feature mapping ϕ , define:

$$K(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^\top \phi(\mathbf{q}) \in \mathbb{R},$$

where $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$ are some points from the data space

- example: for $\phi(\mathbf{x}) = (x_1^2, x_2^2)$, the corresponding kernel K is

$$K(\mathbf{p}, \mathbf{q}) = (p_1^2, p_2^2)^\top (q_1^2, q_2^2) = (p_1 q_1)^2 + (p_2 q_2)^2$$

Non-Linear SVM: Some Popular Kernels

- **polynomial kernel** has the form $K(\mathbf{p}, \mathbf{q}) = (1 + \mathbf{p}^\top \mathbf{q})^P$ for $P \geq 1$
 - it's corresponding ϕ has a dimension of $\ell = \binom{n+P}{P}$
 - e.g., $P = 2, n = 2$, we have

$$\begin{aligned} K(\mathbf{p}, \mathbf{q}) &= (1 + p_1 q_1 + p_2 q_2)^2 \\ &= 1 + (p_1 q_1)^2 + (p_2 q_2)^2 + 2p_1 q_1 + 2p_2 q_2 + 2p_1 p_2 q_1 q_2 \end{aligned}$$

the corresponding mapping is

$$\phi(\mathbf{x}) = (1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1 x_2) \in \mathbb{R}^6$$

where $\binom{4}{2} = 6$.

- **Gaussian kernel** has the form $K(\mathbf{p}, \mathbf{q}) = e^{-\delta \|\mathbf{p} - \mathbf{q}\|_2^2}$ where $\delta > 0$
 - its corresponding ϕ has **infinite dimensions**

The Representer Theorem

- The representer thm. states that the optimal $\mathbf{w}$ to the nonlinear SVM problem is

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \cdot \phi^{(i)} \in \mathbb{R}^\ell$$

where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_m) \in \mathbb{R}^m$, which means $\mathbf{w} \in \mathbb{R}^\ell$ can be parameterized by the m values of α_i 's, even for $\ell \gg m$

- hence, the non-linear SVM problem can be transformed into

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^\ell, \boldsymbol{\alpha} \in \mathbb{R}^m} \quad & \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y^{(i)} \mathbf{w}^\top \phi^{(i)} \geq 1, \quad i = 1, \dots, m, \\ & \mathbf{w} = \sum_{j=1}^m \alpha_j \phi^{(j)} \end{aligned}$$

The Kernel Trick

- substitute the structure of $\mathbf{w}$ into the problem yields

$$\begin{aligned} \min_{\alpha \in \mathbb{R}^m} \quad & \left\| \sum_{j=1}^m \alpha_j \phi^{(j)} \right\|_2^2 \\ \text{s.t.} \quad & y^{(i)} \left(\sum_{j=1}^m \alpha_j \phi^{(j)} \right)^\top \phi^{(i)} \geq 1, \quad i = 1, \dots, m, \end{aligned}$$

- recall that $K(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^\top \phi(\mathbf{q})$, we further introduce the **kernel matrix**:

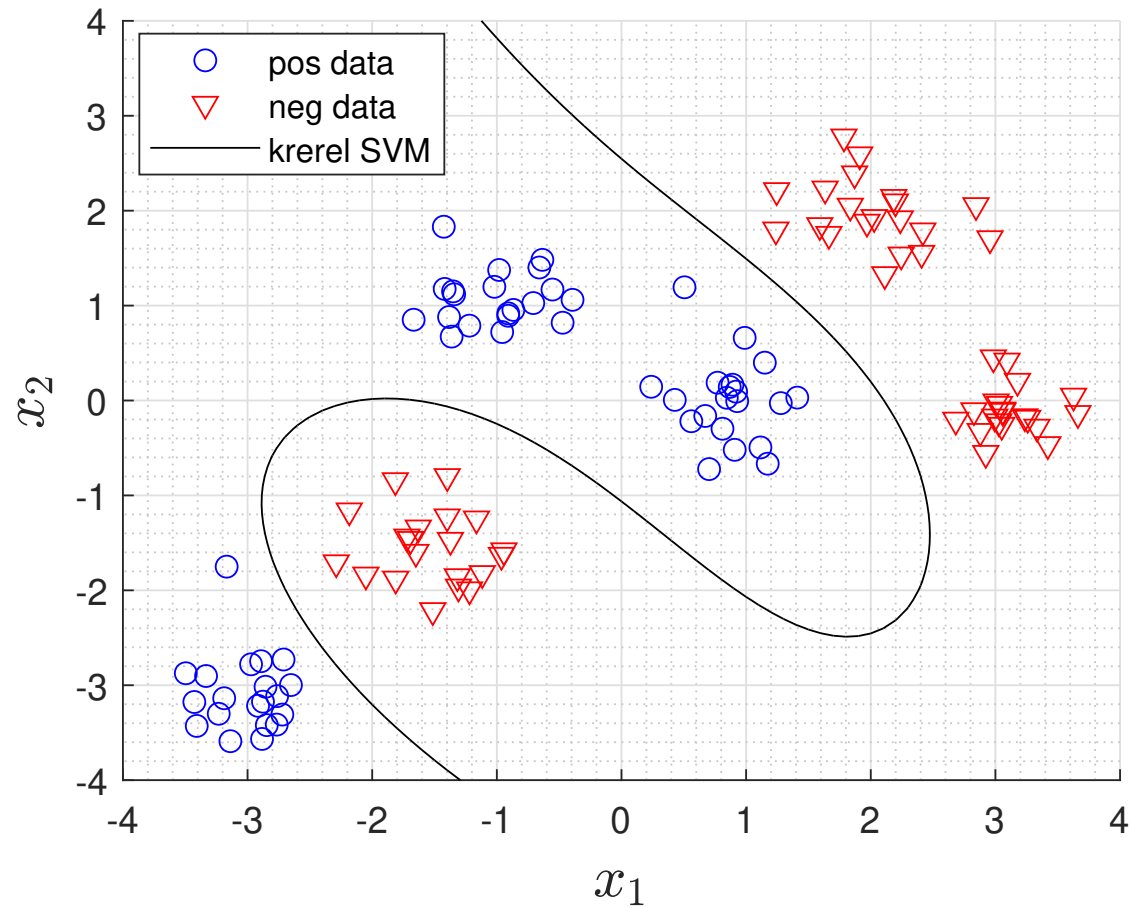
$$\mathbf{K} = \begin{bmatrix} K(\mathbf{x}^{(1)}, \mathbf{x}^{(1)}) & K(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) & \dots & K(\mathbf{x}^{(1)}, \mathbf{x}^{(m)}) \\ K(\mathbf{x}^{(2)}, \mathbf{x}^{(1)}) & K(\mathbf{x}^{(2)}, \mathbf{x}^{(2)}) & \dots & K(\mathbf{x}^{(2)}, \mathbf{x}^{(m)}) \\ \vdots & \vdots & \ddots & \vdots \\ K(\mathbf{x}^{(m)}, \mathbf{x}^{(1)}) & K(\mathbf{x}^{(m)}, \mathbf{x}^{(2)}) & \dots & K(\mathbf{x}^{(m)}, \mathbf{x}^{(m)}) \end{bmatrix}$$

which allows us to simplify the kernel SVM problem into

$$\begin{aligned} \min_{\alpha \in \mathbb{R}^m} \quad & \alpha^\top \mathbf{K} \alpha \\ \text{s.t.} \quad & \mathbf{y} \odot \mathbf{K} \alpha \geq \mathbf{1} \end{aligned}$$

which is just a convex problem with problem size m

Kernel SVM: Toy Demo



- the given toy data is non-separable
- we apply the kernel trick by a Gaussian kernel with $\delta = 0.05$